# Ringel's Conjecture

Walid Elkersh

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# Chapter 1

# Statement

**Definition 1.1.** Ringel's conjecture for all $n$. (Open problem)

**Definition 1.2.** Ringel's conjecture for sufficiently large $n$.

# Chapter 2

# Combinatorial Primitives (Primitives.lean)

**Definition 2.1.** Near-difference rainbow colouring of $K_{2n+1}$.

**Theorem 2.2.** *ND-colouring is a 2-factorization.*

**Lemma 2.3.** $\acute{s}(c+1)$ *step realizes colour $c$.*

**Definition 2.4.** Case predicates.

# Chapter 3

# Tree Structure (TreeStructure.lean)

**Theorem 3.1.** *A tree on* $n$ *vertices has* $n/(4k)$ *leaves or* $n/(4k)$ *vertex-disjoint bare paths of length* $k$*.*

**Theorem 3.2.** *Connects split to case A/B/C division (MPS Lemma 2.2).*

## Chapter 4

# Case Division (CaseDivision.lean)

**Theorem 4.1.** *Routes a tree into Case A, B, or C.*

## Chapter 5

# Case A — Many Leaves (CaseA.lean)

**Definition 5.1.** Signed walk sum.

**Lemma 5.2.** *Tree embedding via signed edge-increment.*

**Lemma 5.3.** *Littlewood–Offord bound for vertex collisions.*

**Lemma 5.4.** *Absorption matching probability.*

**Theorem 5.5.** *Main Case A result.*

## Chapter 6

# Case B — Many Bare Paths (CaseB.lean)

**Lemma 6.1.** *Bare-path embedding.*

**Theorem 6.2.** *Main Case B result.*

# Chapter 7

# Case C — Small Core (CaseC.lean)

**Lemma 7.1.** *Greedy core embedding.*

**Lemma 7.2.** *Leaf extension.*

**Theorem 7.3.** *Main Case C result.*

# Chapter 8

# Probability Infrastructure (ProbBounds.lean)

**Definition 8.1.** Counting measure on finite types.

**Lemma 8.2.** *Positive probability implies existence.*

# Chapter 9

# Proof Assembly (Spine.lean + Proof.lean)

**Theorem 9.1.** *Kotzig cyclic-shift decomposition.*

**Theorem 9.2.** *Rainbow copy exists.*

**Theorem 9.3.** *Full assembly for large $n$.*